

Simplified Simplex: A Complete Walkthrough

Why this problem requires two pivots, not one

We solve the linear program

$$\text{maximize } Z = 100W + 100C$$

$$\text{subject to } W + C \leq 700$$

$$W + 2C \leq 1000$$

$$2W + C \leq 1000, \quad W, C \geq 0$$

Add slack variables so that each constraint is an equality:

$$W + C + s_1 = 700 \quad W + 2C + s_2 = 1000 \quad 2W + C + s_3 = 1000$$

At the starting solution, $W = C = 0$, so $s_1 = 700$, $s_2 = 1000$, $s_3 = 1000$, and $Z = 0$.

Step 1: Choose an entering variable

Both W and C have objective coefficient 100. Either can enter; choose W as a tie-break. Hold $C = 0$ and increase W from zero.

The ratio test asks how far W can increase before each slack becomes negative:

Constraint	When $C = 0$	Bound on W
s_1	$W + s_1 = 700$	$W \leq 700$
s_2	$W + s_2 = 1000$	$W \leq 1000$
s_3	$2W + s_3 = 1000$	$W \leq 500$

The smallest nonnegative ratio is 500. Therefore s_3 reaches zero first: s_3 leaves the basis, and W enters the basis.

After pivot 1

$$W = 500, \quad C = 0, \quad s_1 = 200, \quad s_2 = 500, \quad s_3 = 0$$

$$Z = 100(500) + 100(0) = 50,000$$

Step 2: Check whether the first corner is optimal

This is the crucial check. Solve the binding third constraint for W and substitute it into the objective.

$$2W + C + s_3 = 1000 \text{ implies } W = 500 - (1/2)C - (1/2)s_3$$

$$Z = 100W + 100C = 50,000 + 50C - 50s_3$$

When $s_3 = 0$, increasing C raises Z by 50 for each additional unit of C . Thus $(W, C) = (500, 0)$ is feasible but not optimal: C must enter next.

Step 3: Second ratio test

Keep $s_3 = 0$ and increase C . The equation above gives $W = 500 - (1/2)C$. Substitute this into the two remaining slacks.

$$s_1 = 700 - W - C = 200 - (1/2)C \text{ so } C \leq 400$$

$$s_2 = 1000 - W - 2C = 500 - (3/2)C \text{ so } C \leq 1000/3$$

The smaller bound is $C = 1000/3$. Therefore s_2 reaches zero first: s_2 leaves the basis and C enters the basis.

Step 4: Compute and verify the final solution

$$C = 1000/3$$

$$W = 500 - (1/2)(1000/3) = 1000/3$$

$$s_1 = 700 - 1000/3 - 1000/3 = 100/3, \quad s_2 = 0, \quad s_3 = 0$$

$$Z = 100(1000/3) + 100(1000/3) = 200,000/3 \approx 66,666.67$$

Optimal solution: $W = C = 1000/3 \approx 333.33$, with maximum objective value $Z \approx 66,666.67$.

Stopping rule

The first pivot does not end simplex just because it reaches a corner. After every pivot, check whether a nonbasic decision variable can still improve the objective. Here C had a positive improvement value after pivot 1, so a second pivot was required. At the final point, neither decision variable can be increased while preserving feasibility, so the process stops.